

National Centre for Physics

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Spontaneous Symmetry Breaking and the Higgs Mechanism

Internship Research Report

A study of mass generation in gauge theories, with an application to the tree-level Higgsstrahlung process $e^+e^- \to ZH$, analyzed analytically and computationally with Mathematica and FeynCalc

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Abstract

Part I. This report summarizes an internship project devoted to understanding how elementary particles acquire mass in the Standard Model of particle physics. Starting from the role of symmetry in field theory, we develop the ideas of global and local (gauge) invariance, Noether's theorem, and the necessity of the covariant derivative and gauge fields. We then study the Higgs potential, derive the nonzero vacuum expectation value that signals spontaneous symmetry breaking, and show how expanding the field around this vacuum produces a massive scalar excitation—the Higgs boson—with $m_h^2 = 2\lambda v^2$. The central result is the Higgs mechanism itself: substituting the vacuum value into the gauge-covariant kinetic term generates a gauge-boson mass term $\frac{1}{2}q^2v^2A_\mu A^\mu$ without ever violating gauge invariance. We further discuss the Goldstone theorem, the absorption of the would-be Goldstone boson into the longitudinal polarization of the massive vector boson, and the extension to the electroweak gauge group $SU(2)_L \times U(1)_Y$, which yields the observed masses of the $W^\pm$ and Z bosons while leaving the photon massless. Fermion masses through Yukawa couplings and the experimental discovery of the Higgs boson at the LHC are briefly reviewed. A computational component, implemented in Python, visualizes the Higgs potential and the onset of symmetry breaking; the code is provided in the appendix.

Part II. Part II presents the application project of the internship: Higgsstrahlung, the process $e^+e^- \to ZH$ in which an electron and a positron annihilate into a virtual Z boson that radiates a Higgs boson. Higgsstrahlung is the dominant Higgs-production mechanism at a lepton collider running near $\sqrt{s} \approx 250$ GeV, and it exists only because electroweak symmetry breaking ties the Z boson mass to the Higgs field; studying it at tree level therefore exercises, in one calculation, essentially every part of the electroweak Standard Model I set out to learn. Starting from the Higgs mechanism, I traced how the HZZ and e^+e^-Z interactions arise from the Lagrangian, translated them into Feynman rules, and constructed the single s-channel amplitude for the process. The spin-averaged, polarization-summed squared amplitude was computed both by hand and symbolically with FeynCalc, giving the compact result $\langle|\mathcal{M}|^2\rangle = g^2m_Z^2(g_V^2+g_A^2)\,[\lambda\sin^2\theta+8m_Z^2s]/[8c_W^2v^2(s-m_Z^2)^2]$, from which the differential and total cross sections follow in closed form. Numerically, the total cross section rises steeply from the kinematic threshold at $\sqrt{s} = m_Z + m_H \approx 216$ GeV, reaches its maximum of about 0.24 pb near $\sqrt{s} \approx 245$ GeV, and falls off roughly as $1/s$

at higher energies—the behaviour that motivates the choice of 250 GeV for proposed Higgs factories, and consistent with published tree-level Standard Model expectations. Part II explains the computational pipeline block by block, presents and interprets the resulting energy and angular distributions, and closes with the limitations of the tree-level treatment and the natural next steps.

Acknowledgements

I would like to thank my supervisor, Dr. Ishtiaq Ahmed, for guidance and encouragement throughout this internship, and the National Center of Physics (NCP), Islamabad, for providing the opportunity and resources to pursue this project. I am also grateful to my teachers and colleagues for many helpful discussions.

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Part I

The Higgs Mechanism

Chapter 1

Introduction

1.1 Motivation

The Standard Model of particle physics is a gauge theory: its interactions are dictated by requiring invariance under local symmetry transformations. This principle is spectacularly successful—it produces electromagnetism, the weak force, and the strong force from symmetry alone. But it comes with an apparently fatal flaw: *gauge invariance forbids mass terms for the gauge bosons.* A term such as

$$\frac{1}{2}m^2 A_\mu A^\mu \tag{1.1}$$

is not invariant under a gauge transformation and therefore cannot be written into the Lagrangian by hand. Yet experiment is unambiguous: the $W^\pm$ and Z bosons that mediate the weak interaction are very massive ($m_W \approx 80.4$ GeV, $m_Z \approx 91.2$ GeV), while the photon is massless.

The resolution of this tension is the **Higgs mechanism**, proposed in 1964 by Englert and Brout, by Higgs, and by Guralnik, Hagen and Kibble [1, 2, 3]. A scalar field with a symmetric potential settles into a *non-symmetric* lowest-energy state—a phenomenon called spontaneous symmetry breaking—and its nonzero vacuum expectation value converts gauge-invariant interaction terms into mass terms. The mechanism predicts a new particle, the Higgs boson, which was discovered at CERN's Large Hadron Collider in 2012 with a mass of about 125 GeV [5, 6], completing the particle content of the Standard Model.

1.2 Objectives of the internship

The objectives of this internship were:

1. to build the theoretical prerequisites—classical field theory, symmetries, Noether's

theorem, and gauge invariance—from first principles;

2. to derive, rather than merely quote, the key results of spontaneous symmetry breaking: the vacuum expectation value, the Higgs boson mass, and the gauge-boson mass;

3. to understand the Goldstone theorem and how gauge symmetry evades it;

4. to survey the electroweak theory $SU(2)_L \times U(1)_Y$ and the experimental status of the Higgs boson;

5. to develop a small computational project in Python visualizing the Higgs potential and symmetry breaking.

1.3 Structure of the report

Chapter 2 introduces symmetry in field theory, Noether's theorem, and the distinction between global and local invariance, showing how the covariant derivative and the gauge field arise. Chapter 3 studies the Higgs potential and spontaneous symmetry breaking. Chapter 4 derives the Higgs boson mass and the Higgs mechanism in the $U(1)$ toy model, and discusses the Goldstone theorem and the unitary gauge. Chapter 5 extends the discussion to the electroweak sector and fermion masses, and reviews the LHC discovery. Chapter 6 presents the computational project, and Chapter 7 concludes. Python code is listed in Appendix A.

Chapter 2

Symmetry and Gauge Invariance

2.1 Symmetry in physics

A *symmetry* is a transformation that leaves the physics of a system unchanged: the mathematical description may change, but no observable does. A sphere rotated by any angle looks identical; the laws of mechanics are the same today as tomorrow; shifting an entire experiment three metres to the left changes nothing measurable.

Symmetries divide into two classes:

- **Continuous** symmetries, parametrized by a continuously variable quantity (rotations by any angle, translations by any distance);
- **Discrete** symmetries, allowing only specific transformations (parity, charge conjugation, time reversal).

Noether's theorem, discussed below, applies to continuous symmetries.

A simple example: for a free particle with Lagrangian $L = \frac{1}{2}m\dot{x}^2$, the translation $x \to x + a$ (constant a) leaves $\dot{x}$, and hence L, unchanged. This translational symmetry is responsible for momentum conservation.

2.2 Internal symmetry and the group $U(1)$

The symmetries above act on spacetime. *Internal* symmetries instead act on the fields themselves. Consider a complex scalar field $\phi(x)$ and the phase transformation

$$\phi(x) \longrightarrow e^{i\alpha}\,\phi(x), \tag{2.1}$$

with α constant. The value of the field changes, but the spacetime point does not. Such phase factors $e^{i\alpha}$ form the group $U(1)$, the simplest continuous internal symmetry group.

The physical motivation comes from quantum mechanics: multiplying a wavefunction by a phase leaves all probabilities invariant,

$$|\psi'|^2 = \left(e^{-i\alpha}\psi^*\right)\left(e^{i\alpha}\psi\right) = \psi^*\psi = |\psi|^2. \tag{2.2}$$

Likewise the free Lagrangian density of a complex scalar field,

$$\mathcal{L} = \partial_\mu\phi^*\,\partial^\mu\phi - V(\phi^*\phi), \tag{2.3}$$

is invariant under (2.1), because every ϕ is accompanied by a ϕ^* and the constant phases cancel.

2.3 Noether's theorem

Noether's theorem is one of the most profound results in theoretical physics:

> *Every continuous symmetry of the action corresponds to a conserved current j^μ satisfying $\partial_\mu j^\mu = 0$.*

Writing $j^\mu = (\rho, \mathbf{j})$, the conservation law is the continuity equation

$$\frac{\partial\rho}{\partial t} + \nabla\cdot\mathbf{j} = 0, \tag{2.4}$$

which states that the charge $Q = \int j^0\, d^3x$ can change inside a region only by flowing through its boundary. The classic correspondences are summarized in Table 2.1.

Symmetry	**Conserved quantity**
Time translation	Energy
Space translation	Momentum
Rotation	Angular momentum
Global $U(1)$ phase	Electric charge

Table 2.1: Continuous symmetries and their Noether charges.

For the Higgs story the key entry is the last one: the global $U(1)$ symmetry of the complex scalar field implies a conserved charge, which in electrodynamics is electric charge.

2.4 From global to local symmetry

Now promote the constant phase to a *spacetime-dependent* one:

$$\phi(x) \;\longrightarrow\; e^{i\alpha(x)}\,\phi(x). \tag{2.5}$$

This is called a *local* or *gauge* transformation. Under the global transformation the derivative behaves perfectly,

$$\partial_\mu \phi' = e^{i\alpha}\, \partial_\mu \phi \qquad (\alpha \text{ constant}), \tag{2.6}$$

so the kinetic term in (2.3) is invariant. Under the local transformation, however, the product rule produces an extra piece:

$$\begin{aligned} \partial_\mu \phi' &= \partial_\mu \big(e^{i\alpha(x)}\phi\big) = e^{i\alpha(x)}\partial_\mu \phi + \big(\partial_\mu e^{i\alpha(x)}\big)\,\phi \\ &= e^{i\alpha(x)}\Big(\partial_\mu + i\,\partial_\mu \alpha(x)\Big)\phi. \end{aligned} \tag{2.7}$$

The term $i(\partial_\mu \alpha)\phi$ did not exist before. Because of it, $\partial_\mu \phi$ no longer transforms the same way as ϕ itself, and the kinetic term $\partial_\mu \phi^* \partial^\mu \phi$ is *not* invariant. The ordinary derivative compares the field at neighbouring points, and it "notices" that the phase convention is changing from point to point.

2.5 The covariant derivative and the gauge field

The remarkable resolution is not to abandon local symmetry but to introduce a new field, the **gauge field** $A_\mu(x)$, and define the **covariant derivative**

$$\boxed{D_\mu = \partial_\mu + iqA_\mu} \tag{2.8}$$

where q is the charge of the field. If under the transformation (2.5) the gauge field simultaneously shifts as

$$A_\mu \;\longrightarrow\; A_\mu - \frac{1}{q}\,\partial_\mu \alpha(x), \tag{2.9}$$

then the unwanted term in (2.7) is exactly cancelled and

$$D_\mu \phi \;\longrightarrow\; e^{i\alpha(x)}\, D_\mu \phi, \tag{2.10}$$

i.e. $D_\mu \phi$ transforms just like ϕ. Replacing $\partial_\mu \to D_\mu$ everywhere restores full local gauge invariance of the Lagrangian.

The physical interpretation is profound. We demanded that physics be invariant under local $U(1)$ transformations, and the mathematics replied that this is possible only if a new vector field exists and couples to the scalar field. That field, in electrodynamics, is precisely the electromagnetic four-potential A_μ, and its own gauge-invariant dynamics is supplied by the field-strength term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. In this sense,

> *demanding local gauge symmetry forces the existence of interactions.*

2.6 The mass problem

There is, however, a price. A mass term for the gauge field would read

$$\mathcal{L}_{\text{mass}} = \frac{1}{2} m^2 A_\mu A^\mu. \tag{2.11}$$

Under $A_\mu \to A_\mu - \frac{1}{q}\partial_\mu \alpha$ this is manifestly not invariant. Gauge symmetry therefore *forbids* an explicit gauge-boson mass. For electromagnetism this is welcome—the photon is massless—but the carriers of the weak force, $W^\pm$ and Z, are among the heaviest known particles. The remainder of this report develops the mechanism that resolves this contradiction.

Chapter 3

The Higgs Potential and Spontaneous Symmetry Breaking

3.1 Constructing the potential

We seek the simplest potential for a complex scalar field that (i) respects the $U(1)$ symmetry and (ii) has a stable minimum. Invariance under $\phi \to e^{i\alpha}\phi$ requires the potential to depend on ϕ only through the combination

$$|\phi|^2 = \phi^*\phi, \tag{3.1}$$

since $|e^{i\alpha}\phi|^2 = |\phi|^2$. A term like ϕ^3 is excluded: under the transformation it acquires an uncancelled factor $e^{3i\alpha}$ and would explicitly break the symmetry.

The two lowest allowed terms give the celebrated **Higgs potential**

$$\boxed{V(\phi) = \mu^2 |\phi|^2 + \lambda |\phi|^4 , \qquad \lambda > 0.} \tag{3.2}$$

The quartic coupling must be positive so that $V \to +\infty$ as $|\phi| \to \infty$; otherwise the energy would be unbounded from below and no stable vacuum would exist. The character of the theory then hinges entirely on the sign of μ^2:

- $\mu^2 > 0$: the potential is an ordinary bowl with a unique minimum at $\phi = 0$. The vacuum shares the full symmetry of the Lagrangian; nothing dramatic happens.
- $\mu^2 < 0$: the origin becomes a local *maximum* and the minimum moves out to a nonzero field value. This is the interesting case.

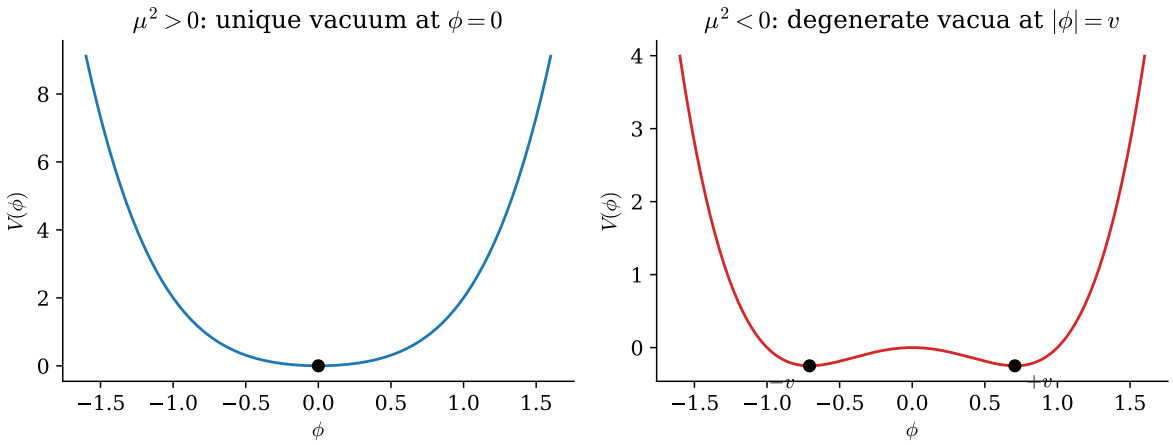


Figure 3.1: One-dimensional slices of the potential $V = \mu^2\phi^2 + \lambda\phi^4$. Left: $\mu^2 > 0$, a unique symmetric vacuum at $\phi = 0$. Right: $\mu^2 < 0$, the origin is unstable and degenerate minima appear at $|\phi| = v$.

3.2 The vacuum expectation value

For $\mu^2 < 0$ write $r = |\phi| \geq 0$, so that

$$V(r) = \mu^2 r^2 + \lambda r^4. \tag{3.3}$$

Stationary points satisfy

$$\frac{dV}{dr} = 2\mu^2 r + 4\lambda r^3 = 2r\left(\mu^2 + 2\lambda r^2\right) = 0. \tag{3.4}$$

The solutions are $r = 0$ and $r^2 = -\mu^2/2\lambda$ (positive, since $\mu^2 < 0$, $\lambda > 0$); the formally negative root is unphysical because a modulus cannot be negative. Evaluating the second derivative shows $r = 0$ is a local maximum—the unstable crown of the "Mexican hat"—while the true minimum lies on the circle

$$|\phi|^2_{\min} = -\frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2}, \qquad \text{i.e.} \qquad \boxed{v = \sqrt{-\frac{\mu^2}{\lambda}}\,.} \tag{3.5}$$

Here we adopt the standard normalization in which the vacuum value of the complex field is $\langle\phi\rangle = v/\sqrt{2}$; the quantity v is the **vacuum expectation value** (VEV).[1]

[1]Conventions differ. If one instead defines v as the minimum of $|\phi|$ directly, one finds $v^2 = -\mu^2/2\lambda$ and correspondingly different numerical factors in derived masses (e.g. $m_h^2 = 8\lambda v^2$ for a potential written as $\mu^2\phi^2 + \lambda\phi^4$ over a real direction). The physics—mass^2 = curvature of the potential at the vacuum—is identical; only the bookkeeping changes. Throughout this report we use the Standard Model convention, in which $m_h^2 = 2\lambda v^2$ and $v \approx 246$ GeV.

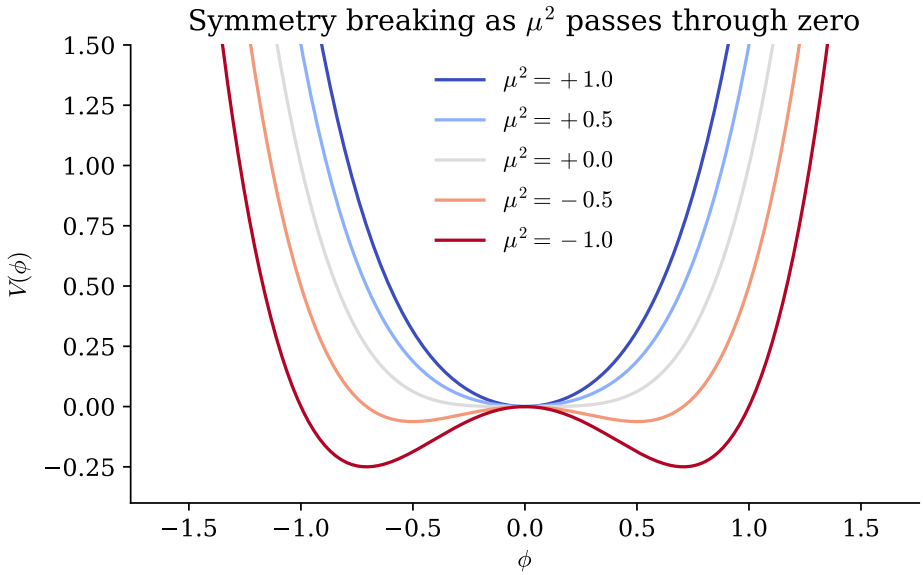


Figure 3.2: Deformation of the potential as μ^2 decreases through zero: the symmetric minimum at the origin smoothly turns into a pair (in the full complex plane, a circle) of degenerate minima.

3.3 Spontaneous symmetry breaking

Equation (3.5) fixes only the *radius* of the vacuum circle, not the angle. Every point on the circle has exactly the same energy, and the equations of motion prefer none of them. Yet the universe must actually *be* in one particular vacuum state, just as a marble balanced on the crown of the hat must eventually roll down and come to rest at one specific point in the valley.

This situation—symmetric equations, non-symmetric ground state—is called **spontaneous symmetry breaking** (SSB). The standard summary is:

> *The Lagrangian possesses the symmetry, but the vacuum does not.*

Two remarks sharpen the idea:

1. If the potential had a single minimum at the origin ($\mu^2 > 0$), the symmetry would still be present but there would be nothing to break: the unique vacuum is itself symmetric. Degeneracy of vacua is essential to SSB.

2. "Vacuum" does not mean "nothing." It is the lowest-energy state of all fields. After SSB the Higgs field takes the constant value $v/\sqrt{2}$ *everywhere in space, even in empty space.* Particles moving through the vacuum are therefore always interacting with this ever-present background—this, and not any picture of "molasses" or "sticking," is the origin of mass. The mass terms arise mathematically from interaction terms evaluated on the shifted vacuum, as we now show.

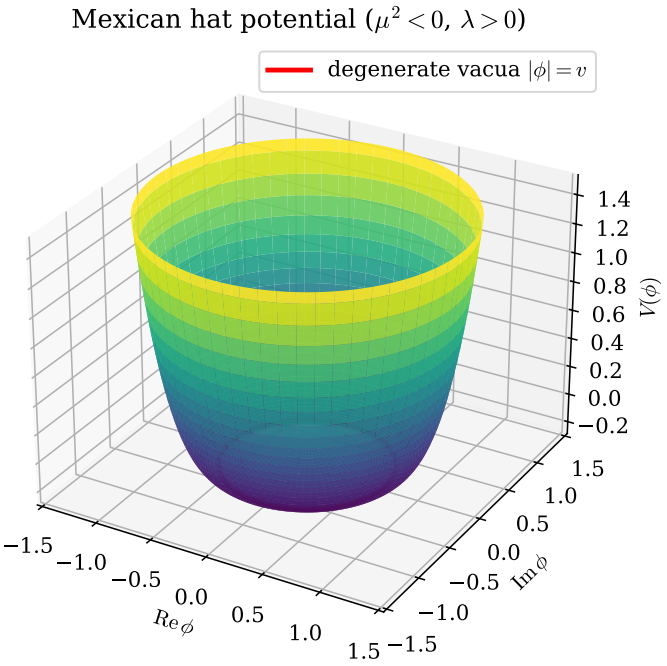


Figure 3.3: The Mexican hat potential over the complex ϕ plane for $\mu^2 < 0$. Every point on the red circle is a minimum of equal energy: the vacuum is infinitely degenerate.

For a *gauge* symmetry there is a further subtlety: two vacua connected by a gauge transformation are not physically different universes but the *same physical state in different gauges*, just as rotating coordinate axes changes a vector's components without changing the vector. Gauge symmetry is a redundancy of description. This observation will explain why no physical massless Goldstone boson survives in the gauged theory.

Chapter 4

The Higgs Boson and the Higgs Mechanism

4.1 Excitations around the vacuum

Particles in quantum field theory are small oscillations of a field about a *stable* configuration. Near a stable minimum any potential is approximately harmonic,

$$V(x) \approx V(x_0) + \tfrac{1}{2}V''(x_0)\,(x - x_0)^2, \tag{4.1}$$

and the curvature $V''(x_0)$ sets the oscillation frequency—which in QFT becomes the particle's mass. Expanding around the unstable point $\phi = 0$ would be meaningless: there $V'' < 0$ and a displacement grows instead of oscillating. We must therefore expand around the true vacuum.

Restrict first to one real direction of the field and write

$$\phi(x) = \frac{1}{\sqrt{2}}\big(v + h(x)\big), \tag{4.2}$$

where $h(x)$ is the fluctuation. Its quanta are **Higgs bosons**: the Higgs *field* fills all space with the constant value $v/\sqrt{2}$, while the Higgs *boson* is a localized ripple on that background—exactly as the photon is a quantum ripple of the electromagnetic field.

4.2 Deriving the Higgs boson mass

Substitute (4.2) into the potential (3.2). Using

$$|\phi|^2 = \frac{(v+h)^2}{2}, \qquad |\phi|^4 = \frac{(v+h)^4}{4}, \tag{4.3}$$

and expanding,

$$\begin{aligned} V &= \frac{\mu^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4 \\ &= \underbrace{\frac{\mu^2 v^2}{2} + \frac{\lambda v^4}{4}}_{\text{vacuum energy}} + \underbrace{\left(\mu^2 v + \lambda v^3\right) h}_{\text{linear term}} + \underbrace{\left(\frac{\mu^2}{2} + \frac{3\lambda v^2}{2}\right) h^2}_{\text{mass term}} + \lambda v\, h^3 + \frac{\lambda}{4} h^4. \end{aligned} \tag{4.4}$$

Consider each piece:

- **Constant:** an overall vacuum energy, irrelevant for particle masses.
- **Linear:** its coefficient is $v(\mu^2 + \lambda v^2)$, which vanishes identically by the minimum condition $\mu^2 = -\lambda v^2$ from (3.5). This had to happen: at the bottom of a valley there is no net force, $dV/d\phi|_v = 0$, so no term linear in the displacement can survive.
- **Quadratic:** using $\mu^2 = -\lambda v^2$,

  $$\left(\frac{\mu^2}{2} + \frac{3\lambda v^2}{2}\right) h^2 = \lambda v^2 h^2 \equiv \frac{1}{2} m_h^2 h^2, \tag{4.5}$$

  giving the **Higgs boson mass**

  $$\boxed{m_h^2 = 2\lambda v^2 = -2\mu^2\ .} \tag{4.6}$$

- **Cubic and quartic:** self-interactions of the Higgs boson, with strengths fixed by the same two parameters μ^2, λ—a genuine prediction of the theory.

The lesson generalizes: *the squared mass of a scalar excitation is the curvature of the potential at the vacuum*, $m_h^2 = V''(v)$ for a canonically normalized field. A shallow valley means a light Higgs; a steep valley means a heavy one. Nothing was inserted by hand—the particle and its mass emerged from expanding a symmetric theory about its asymmetric vacuum.

4.3 The Goldstone theorem

The field ϕ is complex, so (4.2) captured only the radial direction. The general fluctuation about the chosen vacuum has two independent directions (Fig. 4.1):

$$\phi(x) = \frac{1}{\sqrt{2}}\big(v + h(x)\big)\, e^{\,i\theta(x)/v}, \tag{4.7}$$

with $h(x)$ the *radial* mode (climbing the wall of the valley) and $\theta(x)$ the *angular* mode (sliding along the flat valley floor).

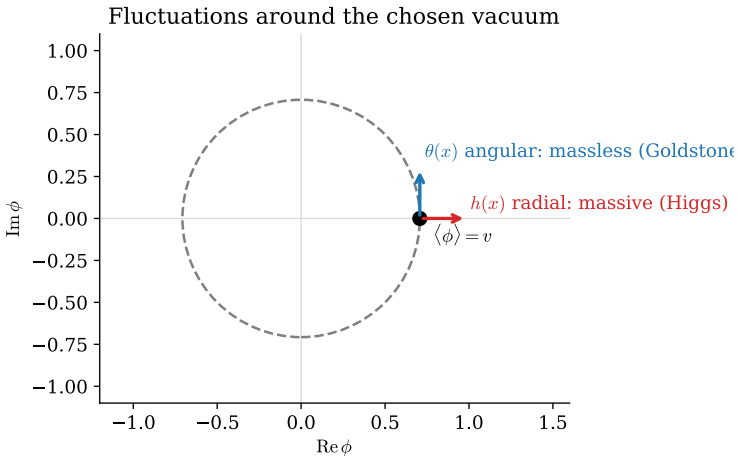


Figure 4.1: Fluctuations about the chosen vacuum. The radial direction climbs the potential and is massive (the Higgs boson h); the angular direction moves along the degenerate circle of vacua at no energy cost and is massless (the would-be Goldstone boson θ).

Because every point of the circle has the same energy, motion in the θ direction costs nothing: the potential contains *no* θ^2 term, and the angular excitation is exactly **massless**. This is a completely general result:

> **Goldstone theorem.** *For every spontaneously broken continuous global symmetry there appears a massless scalar particle—a Goldstone boson.*

For a global symmetry this is the end of the story (such Goldstone modes are physical; the pions of QCD are approximate examples). For the electroweak theory it looks like a disaster: no massless scalar partner of the W and Z has ever been observed. The escape is that the broken symmetry there is not global but *local*.

4.4 The Higgs mechanism in the $U(1)$ model

Now gauge the symmetry: couple ϕ to the gauge field through the covariant derivative (2.8). The gauge-invariant Lagrangian is

$$\mathcal{L} = (D_\mu\phi)^*(D^\mu\phi) - V(\phi^*\phi) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}. \tag{4.8}$$

The kinetic term looks innocent, but since D_μ contains A_μ it hides interactions between the scalar and the gauge field. Insert the vacuum configuration: set $h = \theta = 0$, i.e. $\phi = v/\sqrt{2}$, a constant. Then $\partial_\mu\phi = 0$ and

$$D_\mu\phi = (\partial_\mu + iqA_\mu)\,\frac{v}{\sqrt{2}} = \frac{iqv}{\sqrt{2}}\,A_\mu. \tag{4.9}$$

Therefore

$$(D_\mu\phi)^*(D^\mu\phi) = \left(-\frac{iqv}{\sqrt{2}}A_\mu\right)\left(\frac{iqv}{\sqrt{2}}A^\mu\right) = \frac{q^2v^2}{2}\,A_\mu A^\mu. \tag{4.10}$$

Comparing with the canonical form $\frac{1}{2}m_A^2 A_\mu A^\mu$ we read off

$$\boxed{m_A = q\,v\ .} \tag{4.11}$$

This is the **Higgs mechanism**. We never wrote a forbidden mass term: every term in (4.8) is exactly gauge invariant. The mass *emerged* because the gauge-invariant interaction $(D_\mu\phi)^*(D^\mu\phi)$, evaluated on the nonzero vacuum $\langle\phi\rangle = v/\sqrt{2}$, contains a piece quadratic in A_μ. In one sentence:

> *A gauge-invariant interaction between the Higgs field and the gauge field becomes a gauge-boson mass term once the Higgs field acquires a nonzero vacuum expectation value.*

4.5 Where did the Goldstone boson go? The unitary gauge

What of the massless mode $\theta(x)$ of Section 4.3? Insert the full parametrization (4.7) into (4.8). Because the theory is gauge invariant, we may perform the gauge transformation

$$\phi \to e^{-i\theta(x)/v}\,\phi, \qquad A_\mu \to A_\mu + \frac{1}{qv}\,\partial_\mu\theta, \tag{4.12}$$

which removes $\theta(x)$ from the Lagrangian entirely. This choice is called the **unitary gauge**. The Goldstone boson is not a physical particle at all—it was pure gauge, a redundancy of description, and can be transformed away.

But its degree of freedom is not lost. Count:

	Before SSB	**After SSB**
Complex scalar ϕ	2 (real components)	1 (Higgs boson h)
Gauge field A_μ	2 (transverse polarizations)	3 (transverse + longitudinal)
Total	4	4

A massless vector has two polarizations; a massive one needs three. The extra, longitudinal polarization of the now-massive gauge boson is precisely the degree of freedom of the would-be Goldstone mode. Physicists summarize this as:

> *The gauge boson "eats" the Goldstone boson and becomes heavy.*

Chapter 5

Electroweak Symmetry Breaking

5.1 The electroweak gauge group

The realistic theory replaces $U(1)$ by the electroweak gauge group of Glashow, Weinberg and Salam [4],

$$SU(2)_L \times U(1)_Y, \tag{5.1}$$

with gauge couplings g and g' and four gauge fields: $W_\mu^{1,2,3}$ for $SU(2)_L$ and B_μ for hypercharge $U(1)_Y$. The Higgs field is promoted to a complex $SU(2)$ *doublet*

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \tag{5.2}$$

(four real degrees of freedom) with the same Mexican-hat potential

$$V(\Phi) = \mu^2\, \Phi^\dagger\Phi + \lambda\, (\Phi^\dagger\Phi)^2, \qquad \mu^2 < 0,\ \lambda > 0. \tag{5.3}$$

The minimum again lies at $\Phi^\dagger\Phi = v^2/2$, and in unitary gauge the vacuum may be chosen as

$$\langle\Phi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \tag{5.4}$$

5.2 Masses of the W and Z bosons

Evaluating the doublet kinetic term $|D_\mu\Phi|^2$, with

$$D_\mu = \partial_\mu + ig\,\frac{\sigma^a}{2} W_\mu^a + ig'\frac{Y}{2} B_\mu, \tag{5.5}$$

on this vacuum—the same computation as in Section 4.4, with matrices—yields quadratic terms for three combinations of the four gauge fields:

$$W_\mu^\pm = \frac{W_\mu^1 \mp i W_\mu^2}{\sqrt{2}}, \qquad m_W = \frac{gv}{2}, \tag{5.6}$$

$$Z_\mu = \cos\theta_W\, W_\mu^3 - \sin\theta_W\, B_\mu, \qquad m_Z = \frac{\sqrt{g^2+g'^2}\, v}{2}, \tag{5.7}$$

$$A_\mu = \sin\theta_W\, W_\mu^3 + \cos\theta_W\, B_\mu, \qquad m_\gamma = 0, \tag{5.8}$$

where $\tan\theta_W = g'/g$ defines the weak mixing angle. Three of the four Higgs-doublet degrees of freedom are eaten, giving longitudinal modes to W^+, W^- and Z; the fourth survives as the physical Higgs boson.

The photon remains massless for a structural reason: the particular combination of generators $Q = T^3 + Y/2$ (electric charge) *annihilates the vacuum*. The vacuum is still invariant under the $U(1)_{\rm em}$ generated by Q, so the symmetry-breaking pattern is

$$SU(2)_L \times U(1)_Y \;\longrightarrow\; U(1)_{\rm em}, \tag{5.9}$$

and the gauge boson of the unbroken symmetry—the photon—stays exactly massless. Masslessness of the photon is thus not an accident but a consequence of the surviving symmetry.

5.3 Fermion masses: Yukawa couplings

Gauge invariance also forbids direct fermion mass terms in the Standard Model, because left- and right-handed fermions transform differently under $SU(2)_L$. The Higgs doublet solves this problem too, through gauge-invariant **Yukawa interactions**, e.g. for the electron

$$\mathcal{L}_{\rm Yuk} = -\, y_e\, \bar{L}_e\, \Phi\, e_R + \text{h.c.} \tag{5.10}$$

After SSB, $\Phi \to (0, v/\sqrt{2})^T$, and this interaction becomes a mass term

$$m_e = \frac{y_e\, v}{\sqrt{2}}. \tag{5.11}$$

Every fermion mass in the Standard Model has this form: mass $\propto$ (coupling to the Higgs field) $\times$ (VEV). Heavier particles are simply those that couple more strongly to the Higgs field—a prediction verified at the LHC, where the measured Higgs couplings to t, b, τ and μ scale with their masses.

5.4 Experimental status

The electroweak VEV is fixed by the precisely measured Fermi constant of muon decay:

$$v = \left(\sqrt{2}\, G_F\right)^{-1/2} \approx 246 \text{ GeV}. \tag{5.12}$$

The gauge-boson masses predicted by the mechanism, $m_W \approx 80.4$ GeV and $m_Z \approx 91.2$ GeV, were confirmed at CERN in 1983. The final missing ingredient, the Higgs boson itself, was discovered on 4 July 2012 by the ATLAS and CMS collaborations at the Large Hadron Collider, chiefly in the $H \to \gamma\gamma$ and $H \to ZZ^* \to 4\ell$ channels, with

$$m_h \approx 125 \text{ GeV} \quad \Longrightarrow \quad \lambda = \frac{m_h^2}{2v^2} \approx 0.13. \tag{5.13}$$

All measured properties of the discovered boson—spin 0, even parity, and couplings proportional to mass—are so far consistent with the Standard Model Higgs [5, 6]. The 2013 Nobel Prize in Physics was awarded to Englert and Higgs for the theoretical prediction.

Chapter 6

Computational Project: Visualizing Symmetry Breaking

To complement the analytic work, a small Python project (NumPy + Matplotlib) was developed to visualize the Higgs potential and the onset of spontaneous symmetry breaking. The full source code is given in Appendix A; all figures in this report were produced with it.

The program:

1. plots one-dimensional slices of $V(\phi) = \mu^2\phi^2 + \lambda\phi^4$ for both signs of μ^2 and marks the minima (Fig. 3.1);

2. renders the full Mexican-hat surface over the complex ϕ plane and superimposes the circle of degenerate vacua $|\phi| = v$ (Fig. 3.3);

3. scans μ^2 from positive to negative values, exhibiting the continuous deformation of a symmetric single-well potential into the double-well/hat shape (Fig. 3.2)—a simple picture of what is believed to have happened dynamically in the early universe as it cooled through the electroweak phase transition;

4. illustrates the two fluctuation directions around a chosen vacuum—the massive radial (Higgs) mode and the massless angular (Goldstone) mode (Fig. 4.1).

Beyond producing the figures, the exercise reinforced the central quantitative facts numerically: the minimum of the scanned potentials moves from $r = 0$ to $r = \sqrt{-\mu^2/2\lambda}$ exactly when μ^2 changes sign, and the numerically evaluated curvature V'' at the minimum reproduces the analytic Higgs mass formula.

Chapter 7

Conclusion

This internship traced the complete logical chain behind mass generation in the Standard Model:

symmetry $\rightarrow$ global $U(1) \rightarrow$ local gauge symmetry $\rightarrow$ gauge field & covariant derivative $\rightarrow$ Higgs potential $\rightarrow$ spontaneous symmetry breaking $\rightarrow$ vacuum expectation value $\rightarrow$ expansion around the vacuum $\rightarrow$ Higgs boson & gauge-boson masses.

The key insights are worth restating. Gauge symmetry forbids explicit mass terms, yet demands the very interaction terms that, evaluated on a symmetry-breaking vacuum, *become* mass terms. The Higgs boson is not inserted into the theory; it appears automatically as the radial oscillation of the field about its vacuum, with mass fixed by the curvature of the potential. The dangerous massless Goldstone mode of the broken symmetry is unphysical in a gauge theory: it is absorbed as the longitudinal polarization of the massive vector boson. Extended to $SU(2)_L \times U(1)_Y$, the same mechanism delivers the observed W and Z masses, a massless photon protected by the unbroken electromagnetic symmetry, and fermion masses through Yukawa couplings—all confirmed experimentally, most dramatically by the discovery of the 125 GeV Higgs boson in 2012.

Outlook. Natural continuations of this project include: the Goldstone theorem in full generality and chiral symmetry breaking in QCD; loop corrections to the Higgs potential and the hierarchy problem; the electroweak phase transition in the early universe; and the experimental program of precision Higgs-coupling measurements at the LHC and proposed future colliders, which test whether the discovered boson is exactly the Standard Model Higgs or the first window onto physics beyond it.

Part II

Application Project: Tree-Level Higgsstrahlung $e^+e^- \to ZH$

Chapter 8

Introduction

8.1 Why Higgsstrahlung

The discovery of a Higgs boson at the LHC in 2012 settled the question of whether the electroweak symmetry of the Standard Model is broken by a scalar field, but it opened a more demanding one: is this particle *exactly* the Higgs boson of the Standard Model, or only approximately so? Answering that requires measuring its couplings with a precision that hadron colliders struggle to reach, and the machine repeatedly proposed for the job is an electron–positron collider operating at a centre-of-mass energy near 250 GeV. The reason for that specific number is the subject of this report. In the reaction

$$e^+ + e^- \longrightarrow Z + H, \tag{8.1}$$

known as *Higgsstrahlung* (the Higgs is "radiated" off a Z line, in analogy with bremsstrahlung), the annihilating leptons form a virtual Z boson which then splits into a real Z and a Higgs. The cross section for this process switches on at $\sqrt{s} = m_Z + m_H \approx 216$ GeV, climbs to a maximum of roughly a quarter of a picobarn near 245 GeV, and then dies away as the collider energy is raised further. A "Higgs factory" is simply a machine parked near the top of that curve.

What makes the process a natural internship project is that nothing about it can be understood without the Higgs mechanism itself. The central vertex, HZZ, does not exist in an unbroken gauge theory: its strength is proportional to m_Z^2/v, so it is switched on precisely by the same symmetry breaking that gives the Z its mass. Computing $\sigma(e^+e^- \to ZH)$ is therefore not one more scattering exercise; it is a quantitative test of one's understanding of electroweak symmetry breaking, carried all the way from the Lagrangian to a number in picobarns that experiments actually aim at.

8.2 Objectives

Part I of this report was devoted to the Higgs mechanism itself; this part is its *application project*: having studied how electroweak symmetry breaking works, the task was to make it compute something. Higgsstrahlung requires, on top of that foundation, the machinery of relativistic scattering amplitudes—spinors, Feynman rules, trace techniques, and their implementation in FeynCalc. Concretely, I set out to:

1. understand how spontaneous symmetry breaking generates the gauge boson masses and, in the same stroke, the HZZ interaction;
2. derive the two Feynman rules the process needs—the neutral weak coupling of the electron, and the HZZ vertex—from the electroweak Lagrangian rather than quoting them;
3. construct the tree-level amplitude for $e^+e^- \to ZH$, square it with the standard spin and polarization sums, and reduce the answer to a compact invariant form;
4. obtain the differential and total cross sections in closed form, and implement the entire pipeline symbolically in Mathematica/FeynCalc, treating new features that QED never required: an axial coupling (γ^5 at a vertex), a *massive* vector boson in the final state, and its three physical polarizations;
5. evaluate the cross section numerically across collider energies, reproduce the characteristic threshold–peak–falloff shape, and interpret each feature physically;
6. compare the numbers against published Standard Model tree-level expectations for Higgs factories.

8.3 Scope and organization

Everything in this part is at tree level and in the Standard Model; electrons are treated as massless throughout ($m_e \ll m_Z$), an approximation whose only casualty is the tiny s-channel Higgs-exchange diagram, suppressed by the electron Yukawa coupling. The write-up follows one discipline throughout—physics first, then formalism, then computation, with each layer checked against the one before it. Chapter 9 reviews spontaneous symmetry breaking and traces the origin of every mass and coupling the calculation uses. Chapter 10 converts that Lagrangian information into the Feynman rules and the kinematics of a $2 \to 2$ process with unequal final-state masses. Chapters 11 and 12 contain the amplitude, the squared amplitude, and the cross sections; Chapter 13 explains the FeynCalc implementation block by block; Chapter 14 presents and interprets the numerical results; Chapter 15 discusses the tree-level approximation's limits and possible extensions, and Chapter 16 concludes.

Chapter 9

The Higgs Mechanism and Electroweak Symmetry Breaking

The calculation in this report uses exactly four pieces of electroweak input: the Z mass, the electron's vector and axial couplings to the Z, the HZZ vertex, and the relation between the weak coupling g and the Fermi constant G_F. None of these can simply be postulated; all four are outputs of one mechanism. This chapter assembles that mechanism in the minimal form the project needs.

9.1 The problem: gauge symmetry forbids masses

The electroweak sector of the Standard Model is a gauge theory with group $SU(2)_L \times U(1)_Y$: four massless vector fields, coupled to left-handed fermion doublets and right-handed singlets. A naive mass term for a gauge boson, $\frac{1}{2}M^2 A_\mu A^\mu$, is not invariant under a gauge transformation, and an explicit fermion mass term $-m\bar{\psi}\psi$ mixes left- and right-handed components that transform differently, so it is forbidden as well. Yet the weak bosons are emphatically not massless—the weak force is short-ranged—and neither is the electron. The way out, proposed in the 1960s, is not to break the symmetry in the equations but to let the *ground state* break it: spontaneous symmetry breaking.

9.2 The Higgs field and its potential

One introduces a complex scalar doublet ϕ with the potential

$$V(\phi) = \mu^2\,\phi^\dagger\phi + \lambda\,(\phi^\dagger\phi)^2, \qquad \lambda > 0. \tag{9.1}$$

For $\mu^2 > 0$ this is an ordinary massive scalar and the vacuum sits at $\phi = 0$. The interesting case is $\mu^2 < 0$: the origin becomes a local *maximum*, and the minimum of the famous

"Mexican hat" lies on a circle of nonzero field value,

$$\phi^\dagger\phi\Big|_{\min} = \frac{-\mu^2}{2\lambda} \equiv \frac{v^2}{2}, \tag{9.2}$$

which defines the vacuum expectation value v. The field must choose a point on that circle, and any choice breaks the symmetry—not in the Lagrangian, which remains fully gauge invariant, but in the vacuum. Expanding around the chosen vacuum in unitarity gauge,

$$\phi(x) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \tag{9.3}$$

the single surviving real field $h(x)$ is the physical Higgs boson: the radial oscillation of the field about the rim of the hat. Numerically $v \approx 246$ GeV, a value fixed by the Fermi constant through $v = (\sqrt{2}\,G_F)^{-1/2}$—which is precisely why G_F appears among the numerical inputs of my code.

9.3 Gauge boson masses from the kinetic term

Masses appear when the scalar kinetic term $(D_\mu\phi)^\dagger(D^\mu\phi)$ is evaluated on the vacuum (9.3). The covariant derivative couples ϕ to the four gauge fields, and substituting the vacuum value turns those couplings into quadratic terms:

$$(D_\mu\phi)^\dagger(D^\mu\phi) \;\supset\; \frac{g^2v^2}{4}\,W^+_\mu W^{-\mu} + \frac{(g^2+g'^2)\,v^2}{8}\,Z_\mu Z^\mu, \tag{9.4}$$

from which one reads off

$$m_W = \frac{gv}{2}, \qquad m_Z = \frac{\sqrt{g^2+g'^2}\,v}{2} = \frac{gv}{2\cos\theta_W}, \qquad m_W = m_Z\cos\theta_W. \tag{9.5}$$

Here θ_W is the weak mixing angle that rotates the neutral gauge fields into the physical photon (which stays massless—the vacuum is still invariant under electromagnetism) and the massive Z. The relation $m_W = m_Z\cos\theta_W$ is not an assumption but a *prediction* of the mechanism, and my numerical setup uses it directly to obtain $m_W \approx 79.95$ GeV from the measured m_Z and $\sin^2\theta_W$.

9.4 The HZZ vertex: the coupling this project lives on

The step above used only the constant part v of Eq. (9.3). Keeping the full $v + h(x)$ is where this project begins, because the same kinetic term then reads

$$\frac{(g^2 + g'^2)}{8}\,(v+h)^2\,Z_\mu Z^\mu = \frac{m_Z^2}{2} Z_\mu Z^\mu + \frac{m_Z^2}{v}\,h\,Z_\mu Z^\mu + \frac{m_Z^2}{2v^2}\,h^2 Z_\mu Z^\mu. \tag{9.6}$$

The first term is the Z mass; the second is a three-point interaction coupling one Higgs to two Z bosons, with strength m_Z^2/v fixed entirely by quantities we already know. Reading off the Feynman rule (the conventional factor 2 comes from the two identical ways of attaching the Z legs),

$$\boxed{HZZ \text{ vertex:} \qquad i\,\frac{2m_Z^2}{v}\,g^{\mu\nu} \;=\; i\,\frac{g\,m_Z}{\cos\theta_W}\,g^{\mu\nu},} \tag{9.7}$$

where the second form follows from $m_Z = gv/2c_W$ and is the version often printed in textbooks; my code uses the $2m_Z^2/v$ form and I checked the equivalence by hand. Two features deserve emphasis. First, the vertex is proportional to m_Z^2: *the Higgs couples to the Z because, and only because, the Z acquired its mass from the Higgs field.* In an unbroken theory ($v \to 0$ with $m_Z \to 0$) the vertex disappears and Higgsstrahlung simply does not happen. Second, the vertex carries a bare metric tensor $g^{\mu\nu}$ and no momentum dependence, which will make the amplitude in Chapter 11 remarkably clean.

9.5 The neutral-current coupling of the electron

The other vertex the diagram needs is the electron's coupling to the Z. After symmetry breaking, the neutral-current part of the fermion kinetic term takes the form

$$\mathcal{L}_{\rm NC} = \frac{g}{2\cos\theta_W}\,\bar f\,\gamma^\mu\big(g_V^f - g_A^f\gamma^5\big) f\;Z_\mu, \tag{9.8}$$

so the vertex factor is

$$\boxed{e^+e^-Z \text{ vertex:} \qquad -\,\frac{ig}{2\cos\theta_W}\,\gamma^\mu\big(g_V - g_A\gamma^5\big),} \tag{9.9}$$

with, for the electron,

$$g_V = -\tfrac{1}{2} + 2\sin^2\theta_W \approx -0.0375, \qquad g_A = -\tfrac{1}{2}. \tag{9.10}$$

These numbers are consequences of how the electron's left- and right-handed components sit in $SU(2)_L \times U(1)_Y$: the left-handed electron lives in a doublet (third isospin component $-\frac{1}{2}$, the source of both $-\frac{1}{2}$'s above), the right-handed one in a singlet, and the $2\sin^2\theta_W$ shift is the electromagnetic contamination of the Z through the mixing angle. Two points matter for what follows. The near-cancellation in g_V makes the electron's coupling to the Z almost purely axial—a genuinely electroweak fingerprint that QED could never produce—and the presence of γ^5 in Eq. (9.9) is the technical novelty of this project's traces compared with a pure QED calculation. Since parity is violated by exactly this structure, I had to learn how γ^5 behaves inside traces before trusting the machine to do it for me.

9.6 The weak coupling from the Fermi constant

The last input is the value of g itself. At energies far below m_W, the W-exchange amplitude for muon decay collapses to Fermi's contact interaction, giving the matching condition

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8m_W^2} \qquad \Longrightarrow \qquad g = \sqrt{\frac{8\,m_W^2\,G_F}{\sqrt{2}}} \approx 0.649, \tag{9.11}$$

which is exactly the line `gnum = Sqrt[8*mWnum^2*GFnum/Sqrt[2]]` in my code. I preferred this G_F-based determination over quoting $g = e/\sin\theta_W$ with some running α, because G_F is measured extraordinarily well and the choice keeps every numerical input tied to a single experimental anchor.

With masses, couplings, and the crucial HZZ vertex all traced to one mechanism, the field-theory content of the project is fixed. What remains before any amplitude can be written is bookkeeping: stating the Feynman rules together, and working out the kinematics of a collision that produces two particles of *different* masses—the point where this calculation genuinely departs from familiar QED examples. That is the business of the next chapter.

Chapter 10

Theoretical Framework: Feynman Rules and Kinematics

10.1 The single tree-level diagram

At tree level, and with the electron Yukawa coupling neglected (it is of order $m_e/v \sim 2 \times 10^{-6}$), exactly one diagram contributes to $e^+e^- \to ZH$: the leptons annihilate into an off-shell Z, which converts to the real ZH pair through the vertex (9.7). There is no t-channel diagram, no identical particles in the final state, and no interference to track—a simplicity that felt almost suspicious, and that is worth stating precisely because it is real: a t-channel exchange would need an electron–Z–H or electron–H coupling on an external electron line, and both are Yukawa-suppressed to invisibility.

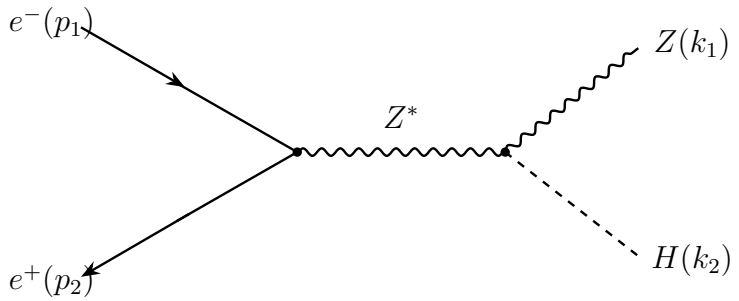


Figure 10.1: The single tree-level (Higgsstrahlung) diagram: s-channel annihilation into a virtual Z, followed by Higgs emission through the HZZ vertex.

10.2 The Feynman rules assembled

Reading the diagram of Fig. 10.1 with the momentum labels $e^-(p_1)\, e^+(p_2) \to Z(k_1)\, H(k_2)$, the ingredients are:

- external legs: $u(p_1)$ for the incoming electron, $\bar{v}(p_2)$ for the incoming positron, and

a conjugated polarization vector $\epsilon^*_\rho(k_1)$ for the outgoing massive Z; the Higgs, being scalar, contributes only the number 1;

- the e^+e^-Z vertex $-\dfrac{ig}{2c_W}\gamma^\mu(g_V - g_A\gamma^5)$ of Eq. (9.9);
- the Z propagator in Feynman gauge, carrying $q = p_1 + p_2$ with $q^2 = s$:

$$\frac{-i\,g_{\mu\nu}}{q^2 - m_Z^2} = \frac{-i\,g_{\mu\nu}}{s - m_Z^2}; \tag{10.1}$$

- the HZZ vertex $i\,(2m_Z^2/v)\,g^{\nu\rho}$ of Eq. (9.7).

Since the collider energies of interest lie far above the Z pole ($s \gtrsim (216\,\text{GeV})^2 \gg m_Z^2$), the propagator denominator never approaches zero and the Z width can be safely omitted from Eq. (10.1)—one of several deliberate tree-level simplifications collected in Chapter 15.

10.3 Two-body kinematics with unequal masses

The central kinematic feature of this process is that the two final-state particles have different, nonzero masses. I set up everything in the centre-of-mass (CM) frame. The beams are massless, so each carries energy and momentum

$$E_{\text{beam}} = |\mathbf{p}| = \frac{\sqrt{s}}{2}. \tag{10.2}$$

Energy–momentum conservation fixes the final-state energies uniquely. Writing $E_Z + E_H = \sqrt{s}$ and $E_Z^2 - E_H^2 = m_Z^2 - m_H^2$ (equal and opposite three-momenta), one solves in two lines:

$$E_Z = \frac{s + m_Z^2 - m_H^2}{2\sqrt{s}}, \qquad E_H = \frac{s + m_H^2 - m_Z^2}{2\sqrt{s}}. \tag{10.3}$$

The common magnitude of the outgoing momenta involves the Källén (triangle) function, which I met here for the first time:

$$\lambda(s, m_Z^2, m_H^2) = s^2 + m_Z^4 + m_H^4 - 2sm_Z^2 - 2sm_H^2 - 2m_Z^2 m_H^2, \qquad |\mathbf{k}| = \frac{\sqrt{\lambda}}{2\sqrt{s}}. \tag{10.4}$$

Its role is transparent at the boundary: λ vanishes when $\sqrt{s} = m_Z + m_H$, so $|\mathbf{k}| = 0$ exactly at threshold,

$$\sqrt{s}_{\,\text{th}} = m_Z + m_H \approx 91.19 + 125.25 = 216.44\ \text{GeV}, \tag{10.5}$$

below which the process is kinematically forbidden. Every feature of the threshold behaviour in Chapter 14 descends from this one function.

With θ the angle between the incoming electron and the outgoing Z, the scalar products among the four momenta follow by direct evaluation in the CM frame ($c_\theta \equiv \cos\theta$):

$$\begin{aligned} p_1{\cdot}p_2 &= \frac{s}{2}, & p_1{\cdot}k_1 &= E_{\rm beam}E_Z - |\mathbf{p}||\mathbf{k}|\,c_\theta, & p_1{\cdot}k_2 &= E_{\rm beam}E_H + |\mathbf{p}||\mathbf{k}|\,c_\theta, \\ p_i^2 &= 0, & p_2{\cdot}k_1 &= E_{\rm beam}E_Z + |\mathbf{p}||\mathbf{k}|\,c_\theta, & p_2{\cdot}k_2 &= E_{\rm beam}E_H - |\mathbf{p}||\mathbf{k}|\,c_\theta, \end{aligned} \tag{10.6}$$

together with the on-shell statements $k_1^2 = m_Z^2$, $k_2^2 = m_H^2$. These eight relations are the complete kinematic dictionary; I chose to parametrize by $\cos\theta$ from the outset because the angular distribution is one of the deliverables. For reference, the Mandelstam variables obey $s + t + u = m_Z^2 + m_H^2$, with $t = (p_1 - k_1)^2 = m_Z^2 - 2p_1{\cdot}k_1$ linear in c_θ.

Chapter 11

The Scattering Amplitude and its Square

11.1 Constructing the amplitude

Reading the diagram against the arrows and inserting the rules of Chapter 10:

$$\mathcal{M} = \Big[\bar{v}(p_2)\,\Big(-\frac{ig}{2c_W}\Big)\gamma^{\mu}(g_V - g_A\gamma^5)\,u(p_1)\Big]\,\frac{-i\,g_{\mu\nu}}{s - m_Z^2}\,\Big[i\,\frac{2m_Z^2}{v}\,g^{\nu\rho}\Big]\,\epsilon^*_{\rho}(k_1). \tag{11.1}$$

The two metric tensors collapse the index chain $\mu \to \nu \to \rho$, and the three explicit factors of $\pm i$ combine to $(-i)(-i)(i) = -i$, leaving the compact result

$$\boxed{\mathcal{M} = -\,\frac{i\,g\,m_Z^2}{v\,c_W\,(s - m_Z^2)}\,\big[\bar{v}(p_2)\,\gamma^{\rho}(g_V - g_A\gamma^5)\,u(p_1)\big]\,\epsilon^*_{\rho}(k_1).} \tag{11.2}$$

This is the expression my code factors out explicitly (the `FCFactorOut` step in Chapter 13), and seeing the machine reproduce the prefactor $-ig\,m_Z^2/[vc_W(s - m_Z^2)]$ was the first handshake between the paper and the computer versions of this calculation. Physically, Eq. (11.2) is one lepton current, one propagator pole at $s = m_Z^2$ (far below our energies), and one factor m_Z^2/v announcing that the whole process is a child of symmetry breaking.

11.2 Spin average and the γ^5 trace

For unpolarized beams I average over the four initial spin configurations and sum over the Z polarizations. The spin sums use the completeness relations for massless spinors, turning the squared current into a trace exactly as in Casimir's trick:

$$\frac{1}{4}\sum_{\text{spins}}\big[\bar{v}\,\Gamma^{\rho}u\big]\big[\bar{v}\,\Gamma^{\sigma}u\big]^* = \frac{1}{4}\,\mathrm{Tr}\big[\not{p}_2\,\gamma^{\rho}(g_V - g_A\gamma^5)\,\not{p}_1\,(g_V + g_A\gamma^5)\gamma^{\sigma}\big]. \tag{11.3}$$

The distinctive ingredient of this trace is γ^5. Multiplying out, the g_V^2 and g_A^2 pieces give the familiar symmetric two-gamma trace structure, while the mixed $g_V g_A$ terms produce a trace of the form $\mathrm{Tr}[\gamma^5 \not{p}_2 \gamma^\rho \not{p}_1 \gamma^\sigma] = -4i\,\epsilon^{\alpha\rho\beta\sigma} p_{2\alpha} p_{1\beta}$—totally *antisymmetric* in $\rho \leftrightarrow \sigma$. That observation does real work in the next step.

11.3 Polarization sum for a massive vector boson

A massive spin-1 particle has three physical polarizations, and their completeness relation differs from the photon's:

$$\sum_{\text{pol}} \epsilon^\rho(k_1)\,\epsilon^{*\sigma}(k_1) = -g^{\rho\sigma} + \frac{k_1^\rho k_1^\sigma}{m_Z^2}. \tag{11.4}$$

The second term is the fingerprint of the longitudinal mode, and it grows with energy—the origin, as we will see, of the dominant behaviour of this process. Because the tensor (11.4) is *symmetric* in $\rho\sigma$, its contraction with the antisymmetric ϵ-tensor piece of Eq. (11.3) vanishes identically. The parity-violating $g_V g_A$ interference therefore drops out of the unpolarized rate, only the combination $g_V^2 + g_A^2$ survives, and the angular distribution comes out symmetric under $\cos\theta \to -\cos\theta$. I found this small mechanism—who kills the γ^5 term, and why—one of the most instructive points of the whole calculation, because it shows that the absence of a forward–backward asymmetry here is not an accident but a structural fact of one diagram with unpolarized beams.

11.4 The result

Carrying out the contractions of Eqs. (11.3) and (11.4) against the kinematic dictionary (10.6)—by hand once, and symbolically in Chapter 13—everything condenses into

$$\boxed{\langle|\mathcal{M}|^2\rangle = \frac{g^2\, m_Z^2\,(g_V^2 + g_A^2)}{8\, c_W^2\, v^2\,(s - m_Z^2)^2}\,\Big[\lambda\,(1 - \cos^2\theta) + 8\, m_Z^2\, s\Big],} \tag{11.5}$$

with $\lambda = \lambda(s, m_Z^2, m_H^2)$ the Källén function of Eq. (10.4). Three checks convinced me the expression is right:

- **Threshold limit.** At $\sqrt{s} \to m_Z + m_H$, $\lambda \to 0$ and the bracket collapses to the isotropic constant $8 m_Z^2 s$: with no momentum available, the decay products cannot remember the beam axis.
- **High-energy limit.** At $s \gg m_Z^2, m_H^2$, $\lambda \to s^2$ and the $\sin^2\theta$ term dominates. Tracing its origin, it comes from the $k^\rho k^\sigma / m_Z^2$ part of Eq. (11.4): the longitudinally polarized Z, whose polarization vector grows like E_Z/m_Z, takes over the process. A

$\sin^2\theta$ distribution is exactly what a longitudinal (helicity-0) vector recoiling against a scalar should produce.

- **Dimensional and structural sanity.** The bracket has dimension GeV^4, the prefactor GeV^{-4}, so $\langle|\mathcal{M}|^2\rangle$ is dimensionless as it must be; and the only angular dependence is even in $\cos\theta$, as argued above.

Chapter 12

Differential and Total Cross Sections

12.1 Differential cross section

For $2 \to 2$ scattering in the CM frame the master formula is

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{k}|}{|\mathbf{p}|}\,\langle|\mathcal{M}|^2\rangle, \tag{12.1}$$

the ratio $|\mathbf{k}|/|\mathbf{p}|$ being the flux-over-phase-space factor that automatically encodes the threshold. Inserting Eq. (11.5) and $|\mathbf{k}|/|\mathbf{p}| = \sqrt{\lambda}/s$:

$$\boxed{\frac{d\sigma}{d\Omega} = \frac{g^2\, m_Z^2\,(g_V^2 + g_A^2)}{512\pi^2\, c_W^2\, v^2\, s^2\,(s - m_Z^2)^2}\,\sqrt{\lambda}\,\Big[\lambda\,(1 - \cos^2\theta) + 8\, m_Z^2\, s\Big].} \tag{12.2}$$

12.2 Total cross section

The azimuthal integration gives 2π, and the polar integral needs only $\int_{-1}^{1}(1 - c^2)\,dc = \frac{4}{3}$ and $\int_{-1}^{1} dc = 2$:

$$\boxed{\sigma(s) = \frac{g^2\, m_Z^2\,(g_V^2 + g_A^2)}{192\pi\, c_W^2\, v^2}\;\frac{\sqrt{\lambda}\,\big[\lambda + 12\, m_Z^2\, s\big]}{s^2\,(s - m_Z^2)^2}.} \tag{12.3}$$

This closed form—identical, term for term, to what the symbolic integration returns in Chapter 13—is the central analytic result of the project. Its anatomy already tells the whole story of Chapter 14:

- the factor $\sqrt{\lambda}$ switches the process on at threshold with the characteristic square-root rise of a two-body phase space;
- the propagator factor $1/(s - m_Z^2)^2$ and the explicit $1/s^2$ suppress the rate as the energy grows, since at large s the numerator behaves as $\sqrt{\lambda}(\lambda + 12 m_Z^2 s) \to s \cdot s^2$, giving $\sigma \sim 1/s$ asymptotically;

- the competition between the rising $\sqrt{\lambda}$ and the falling powers of s must produce a maximum a little above threshold—the Higgs-factory peak;
- the overall strength is set by $(m_Z^2/v)^2 \propto g_{HZZ}^2$: measuring this cross section *is* measuring the Higgs–Z coupling.

Chapter 13

Computational Methodology: Mathematica and FeynCalc

The FeynCalc pipeline assembled here goes beyond a pure QED calculation in three respects: a chiral (γ^5) vertex, a massive external vector boson with its three polarizations, and explicit CM kinematics parametrized by $\cos\theta$ rather than by Mandelstam substitution rules. This chapter walks through the actual program in logical blocks, in the same spirit as before: no line appears without its justification, and the symbolic outputs are checked against the boxed results of Chapters 11 and 12.

13.1 Block 1: initialization and physical constants

```
$FeynCalcStartupMessages = False;
Needs["FeynCalc`"];

mZnum = 91.1876; mHnum = 125.25; vevnum = 246.0;
GFnum = 1.1663787*10^-5; sw2num = 0.23126;
cw2num = 1 - sw2num; cwnum = Sqrt[cw2num];
mWnum = mZnum*cwnum;
gnum = Sqrt[8*mWnum^2*GFnum/Sqrt[2]];
gVnum = -1/2 + 2*sw2num; gAnum = -1/2;
GeVtoPb = 0.3893793656*10^9;
```

The calculation is symbolic throughout; numbers enter only at the very end. This block merely prepares them: the measured m_Z, m_H, v, G_F and $\sin^2\theta_W$, from which the derived quantities follow the chain of Chapter 9—$m_W = m_Z c_W$ (Eq. (9.5)), g from the Fermi-constant matching (Eq. (9.11)), and the electron couplings (Eq. (9.10)). Numerically this gives $c_W \approx 0.877$, $m_W \approx 79.95$ GeV, $g \approx 0.649$ and $g_V \approx -0.0375$. The last line is the unit conversion $1\ \text{GeV}^{-2} = 3.894 \times 10^8$ pb, needed because a cross section computed in

natural units emerges in GeV^{-2} while experiments quote picobarns.

13.2 Block 2: kinematics and the scalar-product dictionary

```
FCClearScalarProducts[];
Ebeam = Sqrt[s]/2; pmag = Sqrt[s]/2;
EZ = (s + mZ^2 - mH^2)/(2*Sqrt[s]);
EH = (s + mH^2 - mZ^2)/(2*Sqrt[s]);
kallen = s^2 + mZ^4 + mH^4 - 2*s*mZ^2 - 2*s*mH^2 - 2*mZ^2*mH^2;
kmag = Sqrt[kallen]/(2*Sqrt[s]);
SP[p1, p1] = 0; SP[p2, p2] = 0;
SP[k1, k1] = mZ^2; SP[k2, k2] = mH^2;
SP[p1, p2] = s/2;
SP[p1, k1] = Ebeam*EZ - pmag*kmag*cth;
SP[p2, k1] = Ebeam*EZ + pmag*kmag*cth;
SP[p1, k2] = Ebeam*EH + pmag*kmag*cth;
SP[p2, k2] = Ebeam*EH - pmag*kmag*cth;
```

`FCClearScalarProducts` wipes any kinematics left over from a previous session—a kernel-hygiene habit institutionalized as the first line of every calculation. The rest is Section 10.3 typed in: the beam variables, the final-state energies (10.3), the Källén function (10.4), and the complete dictionary (10.6) assigned through `SP`. Because every scalar product is taught to the kernel *globally*, all later traces emerge already expressed in s and `cth` $= \cos\theta$, with no separate substitution pass. Choosing $\cos\theta$ instead of t as the working variable was deliberate: the angular distribution is a physics deliverable here.

13.3 Block 3: the Feynman rules as functions

```
vertexEEZ[mu_] := -I*g/(2*cw)*GA[mu].(gV - gA*GA[5]);
vertexHZZ[nu_, rho_] := I*(2*mZ^2/vev)*MT[nu, rho];
propagatorZ[mu_, nu_] := -I*MT[mu, nu]/(s - mZ^2);
```

Each rule of Chapter 10 becomes a small Mathematica function: Eq. (9.9) with its γ^5 entering through `GA[5]`, Eq. (9.7) as a metric tensor `MT` scaled by $2m_Z^2/v$, and the Feynman-gauge propagator (10.1) with q^2 already set to s. Writing the vertices as functions of their Lorentz indices keeps the assembly line below readable and mirrors exactly how the rules are quoted on paper.

13.4 Block 4: assembling and simplifying the amplitude

```
amplitudeRaw =
  SpinorVBar[p2, 0].vertexEEZ[mu].SpinorU[p1, 0]*
   propagatorZ[mu, nu]*vertexHZZ[nu, rho]*
   Pair[Momentum[Polarization[k1, I]], LorentzIndex[rho]];

amplitude = Contract[amplitudeRaw] // DiracSimplify //
   FCFactorOut[#, -I*g*mZ^2/(vev*cw*(s - mZ^2))] &;
```

The first statement is Eq. (11.2) before contraction, written in the same order as one reads the diagram: barred spinor, vertex, spinor—the fermion "sandwich," joined by the noncommutative dot— then propagator, second vertex, and the polarization four-vector of the outgoing Z (the `Pair[Momentum[Polarization[...]]]` construction is FeynCalc's $\epsilon^{\rho}(k_1)$). `Contract` lets the metric tensors collapse the $\mu \to \nu \to \rho$ index chain, `DiracSimplify` tidies the Dirac algebra, and `FCFactorOut` pulls out the physical prefactor $-ig\, m_Z^2/[v c_W (s - m_Z^2)]$—at which point the on-screen expression is literally the boxed amplitude (11.2), which is how I verified this stage.

13.5 Block 5: squaring, spin sums, polarization sums

```
amplitudeCC = ComplexConjugate[amplitude];
msqRaw = amplitude*amplitudeCC;
msqSpins = (1/4)*FermionSpinSum[msqRaw];
msqPols = DoPolarizationSums[msqSpins, k1] // Contract;
msqAveraged = msqPols // DiracSimplify // ExpandScalarProduct // Simplify;
```

This block is the entire content of Chapter 11 in five lines. `ComplexConjugate` builds $\mathcal{M}^*$, reversing the spinor chain and conjugating the couplings. `FermionSpinSum` applies the completeness relations and produces the trace (11.3); the explicit 1/4 is the average over the two spin states of each beam—kept visible, as always, because it is physics rather than algebra. The genuinely new call is `DoPolarizationSums[..., k1]`: knowing from the kinematics that $k_1^2 = m_Z^2 \neq 0$, it applies the *massive* completeness relation (11.4), including the longitudinal $k^{\rho}k^{\sigma}/m_Z^2$ term that a photon calculation never sees. `DiracSimplify` then evaluates the traces (including the γ^5 ones, whose ϵ-tensor pieces duly cancel against the symmetric polarization tensor), `ExpandScalarProduct` distributes dot products of momentum sums so the dictionary of Block 2 can act on every term, and `Simplify` condenses the output.

Result and what it says. The machine returns

$$\langle|\mathcal{M}|^2\rangle = -\frac{g^2 m_Z^2 (g_V^2 + g_A^2)}{8 c_W^2 v^2 (s - m_Z^2)^2}\Big[(c_\theta^2 - 1)\lambda - 8 m_Z^2 s\Big], \tag{13.1}$$

which is Eq. (11.5) with the bracket written in the machine's preferred sign convention—I checked the equivalence by expanding both. Seeing $g_V^2 + g_A^2$ emerge with no $g_V g_A$ cross term was the computational confirmation of the antisymmetry argument of Chapter 11.

13.6 Block 6: cross sections

```
dSigmadOmega = 1/(64*Pi^2*s)*(kmag/pmag)*msqAveraged //
   ExpandScalarProduct // Simplify;
sigmaTotal = 2*Pi*Integrate[dSigmadOmega, {cth, -1, 1}] // Simplify;
```

The first line is the master formula (12.1) verbatim, with `kmag/pmag` the flux-over-phase-space ratio $|\mathbf{k}|/|\mathbf{p}|$. Because `msqAveraged` depends on the angle only through `cth`, the total cross section is a one-dimensional integral: the azimuthal 2π explicit, the polar integration done symbolically. The output is precisely the closed form (12.3)—numerator $\sqrt{\lambda}(\lambda + 12 m_Z^2 s)$ and all—which is the second, independent handshake between paper and machine.

13.7 Block 7: numerical evaluation

```
numericalRules = {g -> gnum, cw -> cwnum, gV -> gVnum, gA -> gAnum,
   vev -> vevnum, mZ -> mZnum, mH -> mHnum};
sigmaPb[sqrtS_?NumericQ] :=
   (sigmaTotal /. numericalRules /. s -> sqrtS^2)*GeVtoPb;
resultTable = Table[{sVal, sigmaPb[sVal]}, {sVal, {250, 350, 500}}];
```

Only now do numbers replace symbols. `numericalRules` maps every symbolic constant to its Block-1 value, `sigmaPb` packages the whole analysis as a single function $\sqrt{s} \mapsto \sigma$ in picobarns (the `?NumericQ` guard keeps it from evaluating on symbolic input), and the final table evaluates the three benchmark energies. The same function generated every curve in Chapter 14, so the figures and the table cannot drift apart—they are one expression evaluated at different points.

Chapter 14

Results and Analysis

14.1 Benchmark values

Table 14.1 collects the total cross section at the three benchmark energies, together with the threshold and the location of the maximum found numerically from the same closed form.

Table 14.1: Tree-level totals from Eq. (12.3) with the inputs of Chapter 13. The peak position was obtained by numerical maximization of the same expression.

$\sqrt{s}$ (GeV)	$\sigma(e^+e^- \to ZH)$ (pb)
250	0.238
350	0.128
500	0.057
threshold: $\sqrt{s}_{\text{th}} = 216.44$ GeV	
maximum: $\sigma = 0.239$ pb at $\sqrt{s} \approx 244.8$ GeV	

14.2 The energy dependence

Figure 14.1 is the central result. Three regimes are visible, and each maps onto one factor of Eq. (12.3).

Threshold behaviour. The rise from zero at 216.44 GeV is governed entirely by $\sqrt{\lambda} \propto |\mathbf{k}|$: just above threshold the outgoing pair moves slowly, the available phase space shrinks to a point, and the cross section switches on like a square root—steep, with (formally) infinite slope at the threshold itself. Figure 14.2 zooms into this region. The bracket $[\lambda + 12m_Z^2 s]$ is dominated there by the isotropic $12m_Z^2 s$ term, consistent with the flat angular distribution discussed below.

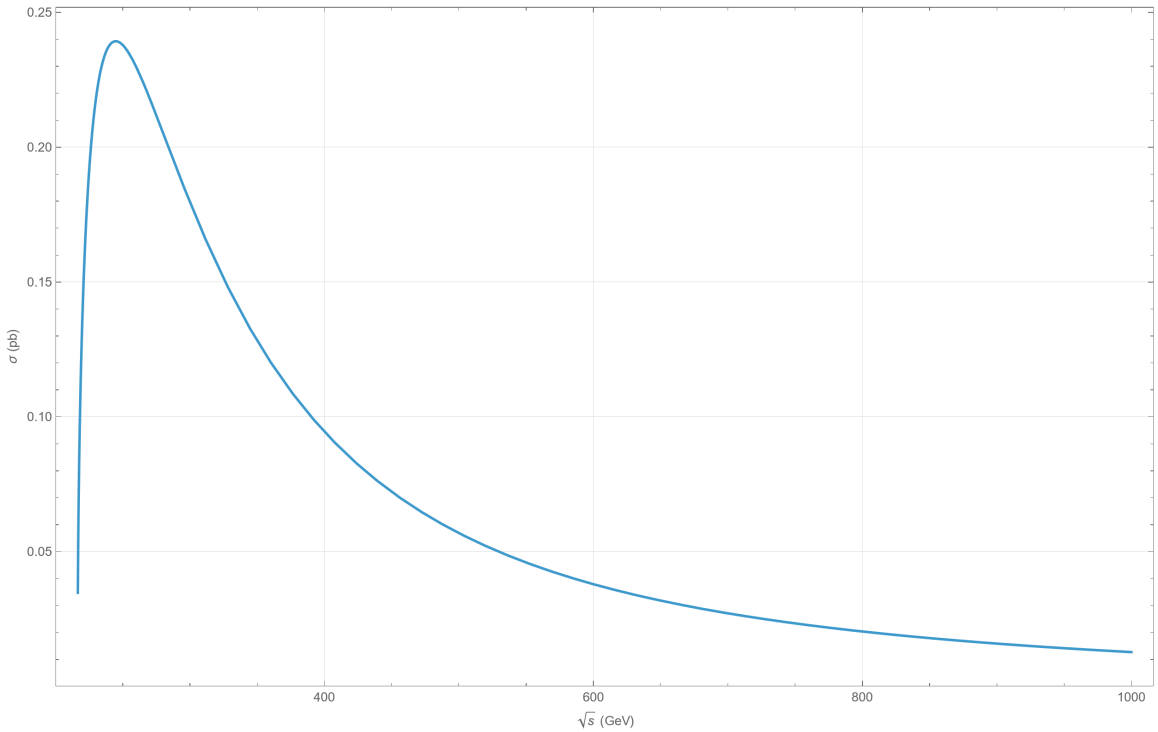


Figure 14.1: Total Higgsstrahlung cross section from threshold to $\sqrt{s} = 1$ TeV, evaluated from the closed form (12.3). The curve rises steeply at $\sqrt{s}_{\text{th}} = 216.4$ GeV, peaks at 0.239 pb near 244.8 GeV, and falls roughly as $1/s$ thereafter.

Why the peak sits near 240–250 GeV. The maximum at $\sqrt{s} \approx 245$ GeV is a competition. Phase space ($\sqrt{\lambda}$) still grows rapidly just above threshold, while the falling factors $1/s^2$ and $1/(s - m_Z^2)^2$ decline smoothly; the product must therefore peak once the square-root growth saturates, which happens within a few tens of GeV of threshold. There is nothing resonant about this maximum—no particle sits at 245 GeV—it is purely the geometry of a two-body phase space opening against a $1/s^3$-ish dynamical decline. This is the quantitative answer to why proposed Higgs factories (ILC, FCC-ee, CEPC) choose $\sqrt{s} \approx 240$–250 GeV: the machine parks where σ is flat and maximal, and my curve shows the plateau is broad —$\sigma(250) = 0.238$ pb is within half a percent of the true peak value.

Why the cross section falls at high energy. Beyond the peak the asymptotics take over: $\sqrt{\lambda}\,(\lambda + 12m_Z^2 s) \to s^3$ while the denominator supplies s^4, so $\sigma \sim 1/s$. Physically, this is the generic fate of an s-channel annihilation through a point-like current: the propagator pays $1/(s - m_Z^2)$ twice, and no compensating growth is available—in contrast, at higher energies the competing WW-fusion channel $e^+e^- \to \nu\bar{\nu}H$, with its t-channel exchange, grows logarithmically and overtakes Higgsstrahlung; computing that crossover is a natural sequel to this project. At $\sqrt{s} = 500$ GeV my curve has already dropped to 0.057 pb, a factor four below the peak, and by 1 TeV to 0.013 pb.

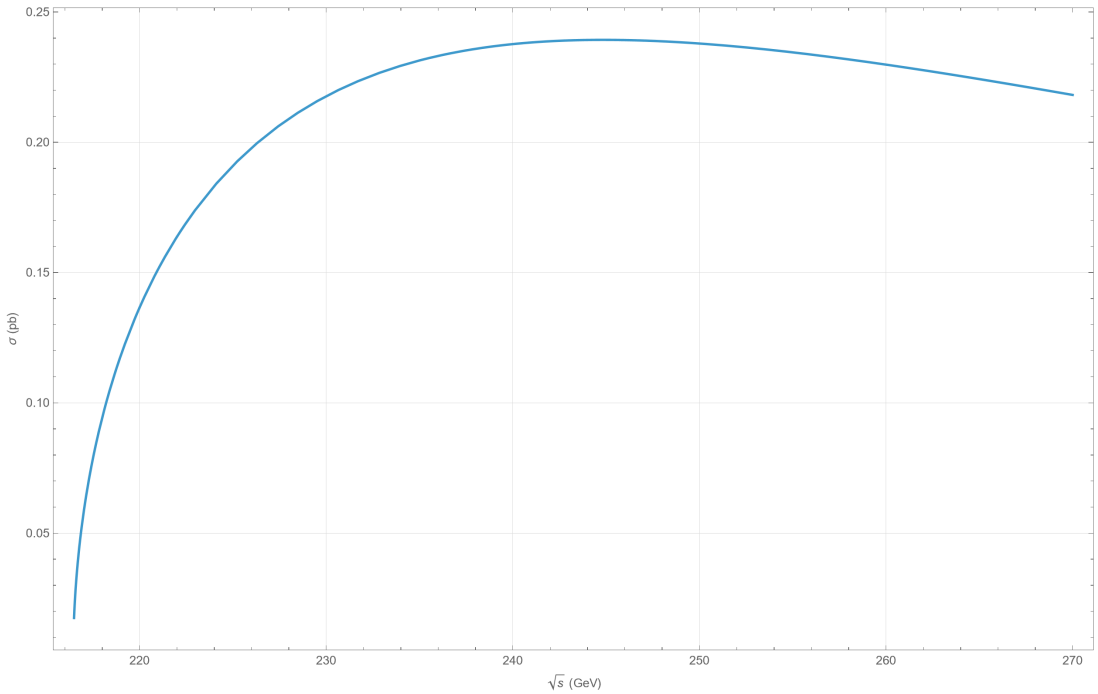


Figure 14.2: The threshold region in detail. The square-root rise imposed by the phase-space factor $\sqrt{\lambda}$ carries the cross section from zero to within a few percent of its maximum in roughly 25 GeV.

14.3 The angular distribution

Figure 14.3 displays $d\sigma/d\Omega$ at the three benchmark energies. Two features carry the physics.

First, every curve is even in $\cos\theta$: the parity-violating $g_V g_A$ interference cancelled against the symmetric polarization sum (Chapter 11), so at tree level with unpolarized beams there is no forward–backward asymmetry for the Z—a nontrivial statement, given that the underlying vertex is almost maximally parity violating ($|g_A| \gg |g_V|$).

Second, the *shape* evolves with energy exactly as Eq. (11.5) dictates. At 250 GeV, close to threshold, the isotropic $8m_Z^2 s$ term still competes with the $\lambda \sin^2\theta$ term and the distribution is only mildly curved: the Z emerges nearly isotropically. By 500 GeV the $\sin^2\theta$ profile dominates, vanishing visibly toward $\cos\theta = \pm 1$. The interpretation is the longitudinal-mode enhancement: the helicity-zero polarization vector scales as E_Z/m_Z, so at high energy the produced Z is predominantly longitudinal, and a longitudinal vector recoiling against a scalar in an s-channel spin-1 state populates $\sin^2\theta$. The angular distribution is thus a polarimeter for the produced Z—and experimentally, since the Z decay products remember its polarization, this shape is measurable.

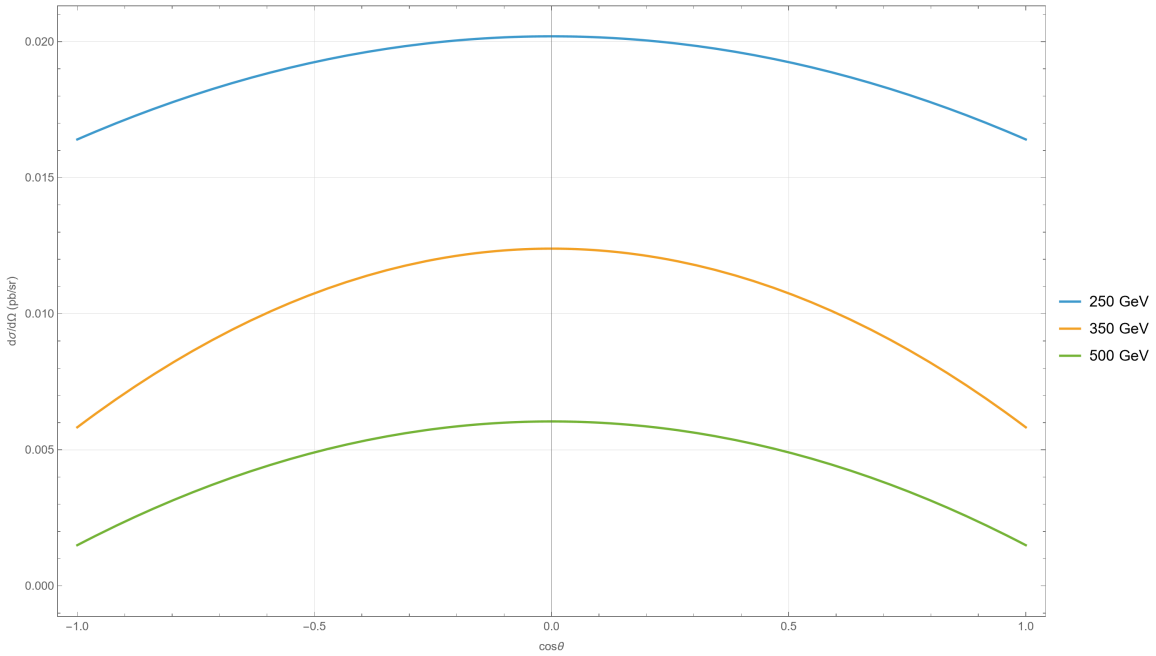


Figure 14.3: Differential cross section $d\sigma/d\Omega$ against $\cos\theta$ at $\sqrt{s} = 250$, 350 and 500 GeV, from Eq. (12.2). All curves are symmetric in $\cos\theta$; the $\sin^2\theta$ component becomes progressively more prominent as the energy grows and the longitudinal Z polarization takes over.

14.4 Comparison with published expectations

The numbers above can be confronted with the Standard Model expectations quoted in the Higgs-factory literature. Unpolarized tree-level studies place $\sigma(e^+e^- \to ZH)$ at roughly 0.2–0.25 pb near $\sqrt{s} = 250$ GeV, with the maximum just below 250 GeV—for example, the ILC physics documentation quotes Higgsstrahlung cross sections of this size as the basis of its 250-GeV programme [5], and the process itself has been the canonical e^+e^- Higgs-production channel since the classic analyses of the 1970s [4]. My values—0.238 pb at 250 GeV, peak 0.239 pb at 244.8 GeV—sit squarely in this range; residual percent-level differences against any specific published table are expected, since such tables typically include initial-state radiation, beam polarization, and electroweak corrections that a strict tree-level, unpolarized calculation deliberately omits.

14.5 The role of the Higgs mechanism, seen from the result

It is worth closing the results chapter by reading Eq. (12.3) once more as a statement about symmetry breaking. The entire rate is proportional to $(m_Z^2/v)^2$—the square of the HZZ coupling that exists only because the Z mass is generated by the Higgs field. If electroweak symmetry were unbroken, $m_Z \to 0$ and the process would vanish identically;

if the observed scalar were not the agent of mass generation, the measured rate would depart from this prediction. That is exactly why the first goal of every proposed Higgs factory is a sub-percent measurement of $\sigma(ZH)$: in the recoil-mass method, tagging only the $Z \to \ell^+\ell^-$ decay determines σ (and hence g_{HZZ}) without touching the Higgs decay at all. The curve of Fig. 14.1 is, in miniature, the physics case for those machines.

Chapter 15

Discussion: Limitations and Extensions

15.1 What the tree-level approximation leaves out

Every simplification made in this project was deliberate, and each defines a correction the calculation does not contain:

1. **Radiative corrections.** One-loop electroweak corrections to $e^+e^- \to ZH$ are known to shift the cross section at the several-percent level—small on the scale of Fig. 14.1, but enormous compared with the sub-percent precision a Higgs factory aims at. A tree-level number is a physics illustration, not a measurement-ready prediction.

2. **Initial-state radiation and beamstrahlung.** Real photon emission off the incoming leptons lowers the effective $\sqrt{s}$ event by event. Near a threshold this matters doubly: it smears the sharp rise of Fig. 14.2 and shifts the optimal operating energy slightly upward—one practical reason 250 GeV is favoured over the naive peak at 245 GeV.

3. **Massless electrons.** Setting $m_e = 0$ removes the Higgs-exchange diagram (Yukawa-suppressed by $(m_e/v)^2 \sim 10^{-11}$ in the rate) and is numerically irreproachable; it is listed here because it is an approximation nonetheless.

4. **Stable final state.** I treated the Z and H as stable outgoing particles. Real analyses fold in the decays ($Z \to \ell\ell$, $q\bar{q}$, $\nu\bar{\nu}$; $H \to b\bar{b}, \dots$), finite widths, and the off-shell tails they generate.

5. **Unpolarized beams.** Linear-collider designs foresee polarized beams, which change both the rate (through the $g_V g_A$ terms that cancelled here) and the achievable precision. My unpolarized 1/4 average is the simplest member of a family.

15.2 Possible extensions

Each limitation above is also a well-posed next project, in roughly increasing order of effort: convolving Eq. (12.3) with an ISR radiator function; repeating the calculation with polarized spinor projectors to recover the $g_V g_A$ physics; attaching the $Z \to \ell^+\ell^-$ decay to study the recoil-mass observable; computing the WW-fusion channel $e^+e^- \to \nu\bar{\nu}H$ and locating the crossover with Higgsstrahlung near $\sqrt{s} \approx 450$ GeV; and, most ambitiously, assembling the one-loop electroweak corrections with modern automated tools. None would require abandoning the framework built here—only extending the same FeynCalc pipeline that this project has, I hope, demonstrated I can operate with understanding rather than on faith.

Chapter 16

Conclusion

This project carried a single electroweak process from the symmetry principles that allow it to a number an experiment could check. Starting from the Higgs mechanism, I traced how one vacuum expectation value simultaneously generates the Z mass, fixes the electron's neutral-current couplings, and creates the HZZ vertex on which Higgsstrahlung lives. The tree-level amplitude turned out to be a single, remarkably clean expression; squaring it required two techniques beyond pure QED—traces with γ^5 and the massive-vector polarization sum—and produced the compact result (11.5), whose structure ($\lambda \sin^2\theta + 8m_Z^2 s$) I could interpret limit by limit. The closed-form cross section (12.3) then explained, factor by factor, the entire phenomenology: the square-root threshold at 216.4 GeV, the broad maximum of 0.24 pb near 245 GeV that defines the Higgs-factory operating point, the $1/s$ decline beyond it, and the growing dominance of the longitudinal Z visible in the angular distribution. The FeynCalc implementation reproduced every boxed formula independently, and its numerical output matches the scale of published Standard Model expectations for unpolarized beams.

Beyond the specific results, the internship consolidated a working method: derive the couplings rather than quote them, keep the symbolic calculation checkable against hand results at every stage, and let the final formula—not the code—explain the physics. The obvious next steps, from initial-state radiation to the WW-fusion crossover, are all within reach of the same machinery.

Appendix A

Python Code for Part I

Listing A.1: `make_figures.py`: generation of all figures in this report.

```
"""Figures for the Higgs mechanism internship report.

Generates:
  fig_potentials.pdf    - V(phi) for mu^2>0 and mu^2<0 (1D slices)
  fig_mexican_hat.pdf   - 3D Mexican hat potential with vacuum circle
  fig_ssb_transition.pdf - potential shape as mu^2 is varied through
     zero
  fig_fluctuation.pdf   - expansion around the vacuum: radial (massive)
     vs angular (massless) directions
"""
import numpy as np
import matplotlib.pyplot as plt
from matplotlib import cm

plt.rcParams.update({
    "font.size": 11,
    "font.family": "serif",
    "mathtext.fontset": "cm",
    "axes.spines.top": False,
    "axes.spines.right": False,
})

lam = 1.0


def V(r, mu2, lam=lam):
    return mu2 * r**2 + lam * r**4


# ---- Figure 1: symmetric vs broken potential (1D slices) ----
fig, axes = plt.subplots(1, 2, figsize=(9, 3.6))
x = np.linspace(-1.6, 1.6, 400)
```

```
31
32  axes[0].plot(x, V(x, mu2=+1.0), color="tab:blue")
33  axes[0].set_title(r"$\mu^2 > 0$: unique vacuum at $\phi=0$")
34  axes[0].plot([0], [0], "ko", ms=6)
35
36  mu2 = -1.0
37  v = np.sqrt(-mu2 / (2 * lam))
38  axes[1].plot(x, V(x, mu2=mu2), color="tab:red")
39  axes[1].set_title(r"$\mu^2 < 0$: degenerate vacua at $|\phi|=v$")
40  axes[1].plot([v, -v], [V(v, mu2), V(v, mu2)], "ko", ms=6)
41  axes[1].annotate(r"$+v$", (v, V(v, mu2)), textcoords="offset points",
42                   xytext=(6, -12))
43  axes[1].annotate(r"$-v$", (-v, V(-v, mu2)), textcoords="offset points",
44                   xytext=(-22, -12))
45
46  for ax in axes:
47      ax.set_xlabel(r"$\phi$")
48      ax.set_ylabel(r"$V(\phi)$")
49  fig.tight_layout()
50  fig.savefig("fig_potentials.pdf")
51  plt.close(fig)
52
53  # ---- Figure 2: 3D Mexican hat ----
54  fig = plt.figure(figsize=(6.5, 5))
55  ax = fig.add_subplot(projection="3d")
56  r = np.linspace(0, 1.35, 120)
57  th = np.linspace(0, 2 * np.pi, 120)
58  R, TH = np.meshgrid(r, th)
59  X, Y = R * np.cos(TH), R * np.sin(TH)
60  Z = V(R, mu2=-1.0)
61  ax.plot_surface(X, Y, Z, cmap=cm.viridis, alpha=0.9, linewidth=0,
62                  antialiased=True, rstride=2, cstride=2)
63  # vacuum circle
64  ax.plot(v * np.cos(th), v * np.sin(th), V(v, -1.0), "r-", lw=2.5,
65          label="degenerate vacua $|\\phi|=v$")
66  ax.set_xlabel(r"$\mathrm{Re}\,\phi$")
67  ax.set_ylabel(r"$\mathrm{Im}\,\phi$")
68  ax.set_zlabel(r"$V(\phi)$")
69  ax.set_title("Mexican hat potential ($\\mu^2<0$, $\\lambda>0$)")
70  ax.legend(loc="upper right")
71  ax.view_init(elev=28, azim=-60)
72  fig.tight_layout()
73  fig.savefig("fig_mexican_hat.pdf")
74  plt.close(fig)
75
76  # ---- Figure 3: transition as mu^2 varies ----
77  fig, ax = plt.subplots(figsize=(6.2, 4))
```

```
for mu2_i, c in zip([1.0, 0.5, 0.0, -0.5, -1.0],
                    plt.cm.coolwarm(np.linspace(0, 1, 5))):
    ax.plot(x, V(x, mu2_i), color=c, label=rf"$\mu^2={mu2_i:+.1f}$")
ax.set_xlabel(r"$\phi$")
ax.set_ylabel(r"$V(\phi)$")
ax.set_title(r"Symmetry breaking as $\mu^2$ passes through zero")
ax.legend(frameon=False)
ax.set_ylim(-0.4, 1.5)
fig.tight_layout()
fig.savefig("fig_ssb_transition.pdf")
plt.close(fig)

# ---- Figure 4: radial vs angular fluctuations ----
fig, ax = plt.subplots(figsize=(6.2, 5), subplot_kw={"aspect": "equal"})
circle = plt.Circle((0, 0), v, fill=False, color="tab:gray", ls="--", lw
    =1.5)
ax.add_patch(circle)
phi0 = np.array([v, 0.0])
ax.plot(*phi0, "ko", ms=8)
ax.annotate(r"$\langle\phi\rangle=v$", phi0, textcoords="offset points",
            xytext=(8, -14))
ax.annotate("", xytext=phi0, xy=phi0 + [0.28, 0.0],
            arrowprops=dict(arrowstyle="->", color="tab:red", lw=2))
ax.text(v + 0.30, 0.05, r"$h(x)$ radial: massive (Higgs)", color="tab:
    red")
ax.annotate("", xytext=phi0, xy=phi0 + [0.0, 0.30],
            arrowprops=dict(arrowstyle="->", color="tab:blue", lw=2))
ax.text(v + 0.03, 0.36, r"$\theta(x)$ angular: massless (Goldstone)",
        color="tab:blue")
ax.set_xlim(-1.2, 1.6)
ax.set_ylim(-1.1, 1.1)
ax.set_xlabel(r"$\mathrm{Re}\,\phi$")
ax.set_ylabel(r"$\mathrm{Im}\,\phi$")
ax.set_title("Fluctuations around the chosen vacuum")
ax.axhline(0, color="0.85", lw=0.8, zorder=0)
ax.axvline(0, color="0.85", lw=0.8, zorder=0)
fig.tight_layout()
fig.savefig("fig_fluctuation.pdf")
plt.close(fig)

print("done")
```

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